

Microcontrollers Quiz #2

Information Engineering (English) 2026

Name:

For real time operation it is necessary to

- ☐ have a processor running as fast as possible at a high clock
- ☒ to have control over what tasks are running and to have short operations (RISC architecture)

The microcontroller stores the variables here:

- ☒ SRAM, this is fast but volatile (is lost when the power is turned off)
- ☐ Flash, it remains there even when the power is turned off

A microcontroller controlling the airbag deployment in a car

- ☐ should have a short development cycle with a fast mockup for the client
- ☒ should be operating in real time with no errors in operation

If I use a bootloader

- ☒ I lose a bit of flash space from the available program memory
- ☐ it will start the program written on the microcontroller every time the board is turned on

GPIO means:

- ☒ General Purpose Input and Output
- ☐ Generic Pin for In and Out operations

Microcontrollers Quiz #2

Information Engineering (English) 2026

Name:

The Arduino concept is to

- ☒ have no peripherals and only connect what is needed as extra modularly with shields
- ☐ have all the peripherals the student might need ready and connected on the PCB

The microcontroller stores the program here:

- ☒ SRAM, this is fast but volatile (is lost when the power is turned off)
- ☐ Flash, it remains there even when the power is turned off

General Purpose Computers

- ☐ were developed far after the microcontrollers were already in embedded systems everywhere
- ☒ were developed about the same time as the first microcontrollers

Far more of these are manufactured each year

- ☒ Microcontrollers
- ☐ General Purpose Computers

Predictable latency means

- ☐ we know the delay accumulated already during the execution of the program
- ☒ we know the max clock cycles before something can get executed

Microcontrollers Quiz #3

Information Engineering (English) 2026

Name:

Instead of `pinMode(13,OUTPUT)`; I could have written

☒ `pinMode(13,1);`

☐ `pinMode(13,0);`

The DDR register:

☒ sets the direction of the individual PINs

☐ sets the value presented on the individual PINs

When I connect a PIN through a switch to GND

☒ unless the switch is pushed there is no value presented to the PIN

☐ when I press it it is 0 when I do not press it it is one

The inputs that Arduino UNO can sense on an external interrupt

☐ HIGH, LOW, RISING, FALLING

☒ CHANGE, LOW, RISING, FALLING

ISR(vector): the vector means:

☐ the name of the function it should jump to

☒ the interrupt vector that called this function

Microcontrollers Quiz #3

Information Engineering (English) 2026

Name:

Instead of `pinMode(13,INPUT)`; I could have written

☐ `pinMode(13,1);`

☒ `pinMode(13,0);`

The PORTB register:

☐ sets the direction of the individual PINs

☒ sets the value presented on the individual PINs

When I connect a PIN through a switch to GND

☐ I can set the PIN to INPUT it will read correctly

☒ I have to set the PIN to INPUT_PULLUP to read correctly

The inputs that Arduino UNO can use as an external interrupt

☒ Digital Pins 2 and 3

☐ All digital pins

`attachInterrupt(pin, ISR, mode)`, the ISR is:

☒ the name of the function it should jump to

☐ the interrupt vector that called this function

Microcontrollers Quiz #4

Information Engineering (English) 2026

Name:

I want to write 2 to the common anode LED 7 segment

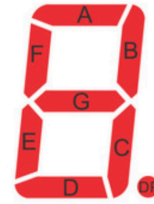
☒ abcdefg. = 0b00100101

☐ abcdefg. = 0b11011010

I want an event every 60 ms (16MHz, prescaler 8):

☒ number of steps is = $16,000,000/8/(1000/60)$

☐ number of steps is = $16,000,000/8/60$



When an overflow event happens

☒ the flag is always set

☐ the interrupt always happens

I want the overflow to happen every 3035 steps

☐ TCNT1= 65535 - 62500;

☒ TCNT1= 62500;

The three events that can happen in a timer:

☐ Overflow, Reached 0, Reached TOP

☒ Overflow, Output Compare A, Output Compare B

Microcontrollers Quiz #4

Information Engineering (English) 2026

Name:

I want to write 2 to the common cathode LED 7 segment

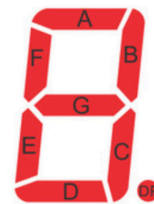
☐ abcdefg. = 0b00100101

☒ abcdefg. = 0b11011010

An event to happen 6000 times/s (16MHz, prescaler 8):

☐ number of steps is = $16,000,000/8/(1000/60)$

☒ number of steps is = $16,000,000/8/6000$



When an output compare event happens

☒ the flag is always set

☐ the interrupt always happens

I want the overflow to happen every 62500 steps

☒ TCNT1= 65535 - 62500;

☐ TCNT1= 62500;

If an interrupt is called while inside an ISR() already

☐ the other interrupt will execute after this one is done

☒ the other interrupt will be lost it will not get called

Microcontrollers Quiz #5

Information Engineering (English) 2026

Name:

CTC mode counting:

☒ CTC counts from 0 to OCR then resets

☐ CTC counts from OCR to OVF

I want to always count to max but have an interrupt at a different value:

☒ this is possible with the OCR interrupt

☐ this is not possible OVF is at the max

COM0A0..1 bit set

☒ what happens with the OCR output at compare (set, clear or toggle)

☐ which PIN is the OCR pin

FOC0A, FOC0B bits

☒ force an OCR event from software

☐ are flags set when OCR happens

Nobel Prize in Physics in 2014 was given for

☐ red LEDs

☒ blue LEDs

Microcontrollers Quiz #5

Information Engineering (English) 2026

Name:

Fast PWM mode counting

☐ Fast PWM counts from 0 to OCR then resets

☒ Fast PWM counts from 0 to OVF like Normal mode

I want to always count to max but have two interrupts at two different values:

☒ this is possible as there are two different OCR interrupts

☐ this is not possible there is only one OCR interrupt

When COM0A0..1 bit are set to clear

☒ OCR event clears, TOV event sets

☐ OCR event sets, TOV event clears

Timer1 Capture event

☐ happens repeatedly when the timer reaches some value

☒ happens when some output pins show a change

Nobel Prize in Physics in 2014 was given for

☐ green LEDs

☒ blue LEDs

Microcontrollers Quiz #6

Information Engineering (English) 2026

Name:

IR light:

- ☒ has longer wavelength compared to visible light
- ☐ has higher frequency compared to visible light

Modulated signals:

- ☒ have a fast carrier component that is modulated by the slow signal component
- ☐ have a slow carrier component that is modulated by the fast signal component

IR communication

- ☐ encodes the 0s and 1s in the frequency of the IR light
- ☒ encodes the 0s and 1s in the timing of the light and dark pulses

UART is

- ☒ asynchron and symmetric
- ☐ synchron and asymmetric

The three events in UART are:

- ☒ RXC, TXC and UDRE
- ☐ DOR, FE, PE

Microcontrollers Quiz #6

Information Engineering (English) 2026

Name:

IR light:

- ☐ has shorter wavelength compared to visible light
- ☒ has lower frequency compared to visible light

Modulated IR light: the signal that modulates

- ☒ changes the amplitude of the carrier (to 0 or 1)
- ☐ changes the frequency of the carrier like in FM radio

RC5 protocol

- ☒ has the same length of pulses for 0 and 1
- ☐ has much longer dark pulse for 1 compared to 0

UART is

- ☐ synchron and asymmetric
- ☒ asynchron and symmetric

The three errors in UART are:

- ☐ RXC, TXC and UDRE
- ☒ DOR, FE, PE

Microcontrollers Quiz #7

Information Engineering (English) 2026

Name:

1W is a protocol that is:

- ☒ asynchron and asymmetric
- ☐ synchron and asymmetric

1W ROM address

- ☒ is a 64 bit unique code written into the device at the factory
- ☐ is a 7 bit code allowing for a maximum of 127 units on the line

1W data transmission happens in time slots

- ☒ always defined by the Master
- ☐ defined by the sender which is either the Master or the slave

For 1W devices it is assumed

- ☒ that the Master has the more precise clock
- ☐ that the Slave has the more precise clock

Error checking in 1W is done

- ☐ by repeating the sent information with bits reversed
- ☒ by sending a CRC - a remainder of a polynomial division

Microcontrollers Quiz #7

Information Engineering (English) 2026

Name:

1W is a protocol that is:

- ☐ asynchron and symmetric
- ☒ asynchron and asymmetric

1W ROM address

- ☐ is a 7 bit code allowing for a maximum of 127 units on the line
- ☒ is a 64 bit unique code written into the device at the factory

1W communication

- ☒ always has 3 steps, reset, rom, ram - and none can be skipped
- ☐ after addressing it no longer needs the reset and the rom command

For 1W devices it is assumed

- ☒ that the Master has the more precise clock
- ☐ that the Slave has the more precise clock

Search ROM needs

- ☒ 64x2 read slots and 64 write slots
- ☐ to go through all possible addresses with brute force

Microcontrollers Quiz #8

Information Engineering (English) 2026

Name:

TWI (I2C) is a protocol that is:

- ☐ asynchron and asymmetric
- ☒ synchron and asymmetric

SLA address

- ☐ is a 64 bit unique code written into the device at the factory
- ☒ is a 7 bit code allowing for a maximum of 127 units on the line

SLA data transmission can happen

- ☒ from Slave to Master if it is a SLA+R
- ☐ only from Master to Slave

A TWI device can

- ☐ initialize a transmission to the Master if an event happened
- ☒ only tell the Master the transmission is too fast by stretching the clock

Error checking in TWI (I2C) is done

- ☐ by repeating the sent information with bits reversed
- ☒ by sending an ACK bit after transmission received

Microcontrollers Quiz #8

Information Engineering (English) 2026

Name:

TWI (I2C) is a protocol that is:

- ☒ synchron and asymmetric
- ☐ asynchron and asymmetric

SLA address

- ☒ is a 7 bit code allowing for a maximum of 127 units on the line
- ☐ is a 64 bit unique code written into the device at the factory

SLA data transmission can happen

- ☒ from Master to Slave if it is a SLA+W
- ☐ only from Slave to Master

If a TWI slave device needs to tell the Master an event happened

- ☐ it can initialize a transmission on the I2C bus
- ☒ it needs to have a separate PIN to tell the Master to ask about it

The ACK bit is

- ☐ always sent by the Master
- ☒ always sent by the receiver, this can be the Master or the Slave

Microcontrollers Quiz #9

Information Engineering (English) 2026

Name:

SPI is a protocol that is:

- ☐ asynchron and asymmetric
- ☒ synchron and asymmetric

SPI address

- ☐ is a 64 bit unique code written into the device at the factory
- ☒ there is no SPI address

SPI data is stable

- ☐ when the CLK is HIGH
- ☒ on one of the clock edges the other being the setup

An SPI device can

- ☐ initialize a transmission to the Master if an event happened
- ☒ not initialize any communication on the SPI lines

Error checking in SPI is done

- ☒ there is no mandatoy error checking in SPI
- ☐ by sending an ACK bit after transmission received

Microcontrollers Quiz #9

Information Engineering (English) 2026

Name:

SPI is a protocol that is:

- ☒ synchron and asymmetric
- ☐ asynchron and asymmetric

SPI address

- ☐ is a 7 bit code allowing for a maximum of 127 units on the line
- ☒ there is no SPI address

SPI data is stable

- ☐ when the CLK is LOW
- ☒ on one of the clock edges the other being the setup

An SPI device can

- ☐ tell the Master the transmission is too fast by stretching the clock
- ☒ not initialize any communication on the SPI lines

Error checking in SPI is done

- ☐ by repeating the sent information with bits reversed
- ☒ there is no mandatoy error checking in SPI

Microcontrollers Quiz #10

Information Engineering (English) 2026

Name:

A usual speed that is used for sampling sound is:

- ☐ 15 kSPS
- ☒ 44.1 kSPS

analogWrite() on an Arduino UNO

- ☒ creates a PWM signal at that output
- ☐ creates a true analog signal at that output

I am converting a 3.3V analog signal on 10 bits, LSB is

- ☐ $3.3 \text{ V} / 10$
- ☒ $3.3 \text{ V} / 1023$

What resolution gives a good quality sound

- ☐ 8 bits
- ☒ 16 bits

Why is a MUX built in to an ADC

- ☐ So it would amplify the signal before converting
- ☒ So it would select the signal - one from multiple inputs before converting

Microcontrollers Quiz #10

Information Engineering (English) 2026

Name:

Studio good quality sample speed for sampling sound is:

- ☐ 48 kSPS
- ☒ 192 kSPS

analogWrite() on an Arduino DUE

- ☐ creates a PWM signal at that output
- ☒ creates a true analog signal at that output

I am converting a 5V analog signal on 12 bits, LSB is

- ☐ $5 \text{ V} / 12$
- ☒ $5 \text{ V} / 4095$

What resolution gives true color

- ☒ 3x8 bits so 24 bits
- ☐ 8 bits

Why is a PGA built in to an ADC

- ☒ So it would amplify the signal before converting
- ☐ So it would select the signal - one from multiple inputs before converting

Microcontrollers Quiz #11

Information Engineering (English) 2026

Name:

A Brushed Motor:

- ☒ loses speed do to the friction that needs to be present at the contacts
- ☐ does not create so much electromagnetic noise as a brushless motor

A Brushless Motor:

- ☒ is easier to cool as the coils are in the stator
- ☐ is easier to drive as it needs only a power voltage

We do not power motors from GPIO directly because

- ☐ the GPIO pin cannot give high enough voltage
- ☒ the GPIO pin cannot give high enough amps

For a DC motor the PWM duty cycle sets

- ☒ the speed at which it is rotating
- ☐ the angle of the rotating shaft

In PID control our input is proportional (integral of, derivative of)

- ☐ the quantity we want to set
- ☒ the error - the disance between the value we want to set and where it is now

Microcontrollers Quiz #11

Information Engineering (English) 2026

Name:

A Brushless Motor:

- ☐ loses speed do to the friction that needs to be present at the contacts
- ☒ does not create so much electromagnetic noise as a brushed motor

A Brushed Motor:

- ☐ is easier to cool as the coils are in the stator
- ☒ is easier to drive as it needs only a power voltage

In a motor driving circuit the diode is present

- ☒ to short the voltage appearing when the motor is turned off
- ☐ to rectify the voltage for the motor

For a servo motor the PWM duty cycle sets

- ☐ the speed at which it is rotating
- ☒ the angle of the rotating shaft

The PWM frequency is set

- ☐ as high as possible
- ☒ as high as the inertia of the driven system demands it to be

Microcontrollers Quiz #12

Information Engineering (English) 2026

Name:

A Node MCU ADC:

- ☒ is limited to 0..1V but can be extended to 0...3.3V with resistors
- ☐ has multiple PINs as inputs that a multiplexer can choose from

A Node MCU ADC

- ☐ is less noisy than the Arduino ADC because the chip is faster
- ☒ is more noisy than the Arduino ADC because the wifi is there

Timing things on a Node MCU

- ☐ is done with `delay()` functions even if the delay is long
- ☒ is better done with `millis()` to not block the main loop

The Node MCU

- ☐ has two sleep modes, active and deep sleep
- ☒ has four sleep modes, active, modem-sleep, light sleep and deep sleep

MQTT

- ☐ requires nodes to stay online
- ☒ can deal with unreliability even with a node falling off the network (last will and testament)

Microcontrollers Quiz #12

Information Engineering (English) 2026

Name:

A Node MCU ADC:

- ☒ has a single PIN and if we need to do multiplexing we need an external chip
- ☐ has multiple PINs as inputs that a multiplexer can choose from

A Node MCU ADC

- ☒ is the same resolution, 10 bits like the Arduino ADC
- ☐ is higher resolution than the Arduino ADC

For a Node MCU

- ☐ if is ok if the main thread is not doing anything for a long while
- ☒ it is important the main loop does not stay idle for long and does not trigger watchdog reset

The Node MCU

- ☐ needs to be awoken by an outside signal once it goes into sleep mode
- ☒ can wake itself up with a PIN change that resets the whole system

MQTT

- ☐ relies on persistent connections
- ☒ does not even need the sender and receiver to be online at the same time